3(a)(i)	Al_2O_3 / aluminium oxide	**1**

3(a)(ii)	• unburned hydrocarbons • react with NO / NO_2 • (to) produce PAN (– a component of photochemical smog) Any 2 points = 1 mark, all 3 points = 2 marks	**2**

Examiner report missing
9701_s25_qp_23 Question 3

3(a)(i)		**3**

element	C (graphite)	Sn
state and appearance	grey shiny solid	silvery solid
electrical conductivity	● good/ conductor	good
type of bonding	● covalent	metallic
type of structure	giant	● giant

3(b)(i)	$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$	**1**

3(b)(ii)	thermal stability **increases** down the group	**1**

3(b)(iii)	**M1** NH_4^+ / ammonium ion / ammonium cation is a proton / $\mathbf{H^+}$ donor **M2** OH^- is a proton / $\mathbf{H^+}$ acceptor **OR** the OH^- / NaOH is a stronger base than NH_3	**2**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as MgCl and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiCl_4 is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid SiCl_4 as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiCl_4 molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and SiCl_4 and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

3(a)(i)	(natural =)lightning	**1**
	(man-made =)internal combustion engines	**1**
3(a)(ii)	$2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$ **OR** $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$	**1**
3(a)(iii)	It / NO_2 reacts with (unburned) hydrocarbons / VOCs **ALLOW** reaction of unburned hydrocarbons / VOCs in presence of NO_2	**1**
3(b)(i)	(+)5 / V	**1**
3(b)(iii)	(NH_3 is an) H^+ acceptor	**1**
3(f)(i)	**M1** one sigma / σ bond **AND** head-on (overlap of) p / sp (orbitals)	**1**
	M2 two pi / π bond(s) **AND** side-on (overlap of / involving) p (orbitals)	**1**
3(f)(ii)	P≡P is much weaker so P_2 is more reactive (than N_2)	**1**

Question 3

(a) (i) Many candidate answers referred to lightning as responsible for one natural occurrence of NO_2. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

(ii) Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of NO_2 as a catalyst in the production of sulfur-based acid rain.

(iii) Candidates who were familiar with this area of the syllabus described how PAN forms from NO_2. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with NO_2.

(iv) Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of H_2 rather than H_2O.

(v) Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

(b) (i) Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

(ii) The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

(iii) Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

(iv) Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

(c) (i) Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

(ii) A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

(d) Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

(e) Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

(f) (i) This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in P_2 identified as sp^2 or sp^3
- in P_2 overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

(ii) Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

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(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when $SiCl_4$ is added to water.

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(f) Many candidates gave the correct answer.

1(a)(i)	**M1** (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond **M2** (N_2 molecules /molecule(s) of N_2) are non-polar	**2**

Question 1

This question assessed inorganic chemistry, with some aspects of physical chemistry. This tested principally in the AO1 category. Candidates showed some facility of recall but were equally divided in response when asked to apply their knowledge to unfamiliar circumstances.

(a)(i) This standard recall question could be answered with easy reference to syllabus points.

(ii) With reference to the syllabus, orbital overlap can be directly quoted; knowledge of the bonding in N_2 is also straightforward, so the number of σ and π bonds should be readily accessible.

(b)(i) Many candidates answered this question well.

(ii) This question gave rise to many variations of answer. Candidates are advised to be brief with their answers: often, correct responses were contradicted by (incorrect) explanations of valence electrons.

(c)(i) Many candidates answered this question well.

(ii) The reaction of P_4O_{10} was better known than that of Al_2O_3. The latter is not less common but requires better honed balancing skills if constructed rather than written in recall.

(d)(i) Many candidates answered this question well.

(ii) Many candidates were able to identify the type of reaction that occurs.

2(a)	58% / no change / stays the same **AND** catalysts don't affect yield / iron is a catalyst	**1**

2(d)(i)	**M1** catalyst **M2** *2 equations to show the catalytic activity of NO_2 in formation of SO_3 from SO_2* • $SO_2 + NO_2 \rightarrow NO + SO_3$ • $2NO + O_2 \rightarrow 2NO_2$ **M3** *describe production of H_2SO_4 from SO_3 and water (can be shown in an equation).* SO_3 then reacts with (rain)water / moisture / water vapour to make sulfuric acid / H_2SO_4 OR $SO_3 + H_2O \rightarrow H_2SO_4$	**3**

2(d)(ii)	**M1** unburned hydrocarbons combine with NO and / NO_2 **M2** to form PAN / peroxyac(et)yl nitrate	**2**

Question 2

Although **(d)** focused on inorganic chemistry, the majority of this question was based on equilibria and rates. Better performing candidates were able to discern the differing aspects here: rates and collision theory formed one notion; equilibria, yield and Le Chatelier's principle the other.

(a) This was generally well answered, with many candidates identifying the null effect of a catalyst on equilibrium position. The best answers stated this explicitly.

(b)(i) This was well answered by candidates, who could link lower yield with the exothermic reaction. Precision of wording was key here.

(ii) Many candidates attempted to answer this rates question in terms of equilibrium yield and were unable to gain full credit as a result. If an item pairs rate and yield within a question, candidates are advised to read the stem of the question carefully and to give relevant answers.

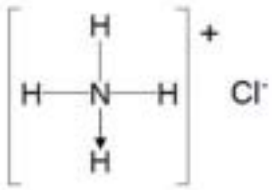
(c)(i) Many candidates were unable sufficiently to distinguish between K_c and K_p. Square brackets must not be used in K_p expressions to avoid confusion with concentration variables. Partial pressure variables can be given as p_X, P_X, $p(X)$, $P(X)$ or $pp(X)$.

(ii) Many numerical answers were correct, though common errors included not adjusting the equilibrium concentrations of N_2 and H_2, or not accounting for **2**NH_3 or **3**H_2 in the stoichiometric equation.

(iii) This was generally well answered by candidates, although some errors of powers of ten were seen in a minority of responses.

(d)(i) Candidates' knowledge of this nitrogen chemistry seemed largely secure, though correct quotation of three equations (two showing NO_2 catalysis and one of the conversion of SO_3 to acid rain) were needed to gain full credit, alongside a statement of the role of NO_x.

(ii) There were many good partial answers to this question, with most referring to PAN but with less precision on how this is produced.

4(a)	 **OR** Correct representation or label showing **M1** (ionic bond between) NH_4^+ and Cl^- **AND** 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 **M2** 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons1	**2**
4(b)(i)	acid–base reaction	**1**
4(b)(ii)	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$	**1**

Question 4

Many individual parts of this question were answered well. Parts **(a)**, **(c)(i)**, **(e)(ii)** and **(f)(ii)** were clearly the most challenging, often as candidates were required to apply several independent parts of their knowledge of the syllabus.

(a) Candidates were not strictly required to give dot-and-cross diagrams, though these and modified Lewis diagrams often produced the clearest responses. Many answers incorrectly showed a pentavalent nitrogen atom — a kind of NH_5 species, with little understanding of the NH_4^+ Cl^- superstructure. The dative bond needed to be clearly distinct from other covalent bonds and ionic charges needed to be given.

(b)(i) There were many correct answers to this question.

(ii) Different versions of this answer were offered, but a large number were a simple addition of NH_4^+ and OH^-, which did not gain credit.

(c)(i) This was well answered by candidates whose knowledge extended to terminology associated with electron-pair donors.

(ii) This was found to be the most challenging item on the paper which required candidates to assess either the basicity or nucleophilicity of each species. Opposing answers were therefore acceptable with enough reasoning.

(d) This was generally well answered by candidates, but the single-step nature of the S_N2 mechanism needed to be clearly evidenced by one transition state; the slower rate of this reaction by a higher energy of the TS relative to either produced in S_N1. Candidates frequently accessed the former part but often then erroneously showed some kind of catalysed reaction with a lower activation energy.

(e)(i) This question was not generally well answered, as candidates were not able to associate electron movement with the correct stage of the reaction and often were very imprecise with their positioning of curly arrows. Many answers had the arrowhead and tail reversed, with a lot of responses suggesting electron movement *from* extra H^+ species.

(ii) This question was reasonably well answered.

(iii) This question was generally well answered, although stray vowels or unspecific/incorrect numbering were observed. Candidates are reminded that the longest carbon chain is the important root of a molecule's name.

(f) (i) Candidates were able to provide a correct answer here by analogy of the 2-bromobutylium species on the left-hand side.

(ii) This was found to be a demanding question, although candidates who recognised the necessary reference to the stability of the intermediate were able to give a reasonably straightforward AO1 account for full credit.

3(a)	strong triple bond / high triple bond enthalpy	**1**
	non-polar (molecule)	**1**
3(b)(i)	*e.g.* $N_2 + 1½O_2 \rightarrow NO + NO_2$	**1**
3(b)(ii)	HNO_2 and HNO_3	**1**
3(b)(iii)	photochemical smog	**1**

Question 3

This question tested candidates' abilities to apply their knowledge in slightly unfamiliar contexts. In-depth knowledge and recall of nitrogen chemistry was crucial, as was the ability to draw clear, labelled, unambiguous diagrams.

(a) Different types of answer were needed to gain full credit. A strong triple bond and high E_a are in effect making the same point.

(b) (i) There were many correct answers here, with balancing being the main difficulty.

(ii) Very few answers correctly referenced both acidic species, HNO_3 and HNO_2. The latter was rarely seen.

(iii) Candidates deployed a full range of answers referring to environmental issues: specifically, this was looking at photochemical smog.

(c) (i) Candidates found it challenging to apply the definition of standard enthalpy change of formation; attempting to conceive of two numbers that would sum to –46 (or –92), or that combined with the ΔH_f would sum to zero.

(ii) Many candidates were able to give good definitions here.

(iii) Candidates did not perform well on this question. Despite '**rate**' being in bold in the question, the large majority of responses attempted to answer in terms of yield and equilibrium position. Nevertheless, some credit was available for correct and non-contradicted information about a decrease in rate. Where collision theory is referenced, candidates are reminded of the need to discuss the *frequency* (i.e., number *per unit time*) of collisions, rather than simply the number.

(d) (i) Many diagrams showed a triple bond between the nitrogen atoms or miscounted the number of non-bonding electrons.

(ii) Candidates did not show a good understanding of hybridisation as responses divided reasonably equally between those who offered sp, sp^2 and sp^3 as answers.

(iii) Some good answers were seen here, but many answers could not be awarded full credit. *Sideways* overlap of p orbitals was needed in the response, or a clear and labelled diagram showing this as part of an unambiguous representation of a pi bond.

2(a)	curved line from (0,0) to reach same horizontal line	**1**
2(d)	*natural process:* lightning	**1**
	man-made process: internal combustion engines	**1**
2(e)(i)	species with 1 or more unpaired electrons	**1**
2(e)(ii)	$NO_2 + SO_2 \rightarrow SO_3 + NO$	**1**
	$NO + \frac{1}{2} O_2 \rightarrow NO_2$	**1**
2(e)(iii)	(formation of) acid rain	**1**

Question 2

(a) Some candidates appeared to confuse graphs of rate versus time with those showing concentration versus time.

(b) Well-rehearsed arguments related the significance of increasing temperature on the position of equilibrium to the example given. More able candidates showed no ambiguity and made explicit reference to the effect of an increase in temperature on the position of this equilibrium in terms of either the '*forward reaction is exothermic'* or '*the reverse reaction is endothermic'*. Less precise answers included '*the reaction moves backwards as it's an exothermic reaction'*.

(c) This question proved to be demanding. A few candidates found K_p using an appropriate method; they worked out the mole fraction of the components, converted them into partial pressures and then substituted these values into the correct expression for K_p. Weaker candidates calculated K_p directly from moles and included moles in the units. Deduction of units also presented difficulties. Many answers gave units in terms of Pa with no attempt to convert kPa to Pa within the calculation.

(d) Lightning was well known as a natural process responsible for formation of atmospheric NO and NO_2. 'Catalytic converter' as the man-made process was a common incorrect answer.

(e) (i) Many correct definitions of the term free radical were seen. Ambiguous descriptions referred to '*unpaired electron'* as '*unshared*' or '*unbonded*' or '*free*'. Common incorrect answers referred to a free radical as *'a species with a lone pair of electrons'*.

(ii) The pair of equations which showed the catalytic role of NO_2 in the oxidation of SO_2 were not well known. The equation to represent the regeneration of NO_2 appeared to cause most difficulties with many incorrectly balanced equations seen.

(iii) The environmental consequence of the oxidation of sulfur dioxide was well known.

(f) (i) Identification of a magnesium containing base was relatively common. Incorrect answers included salts, rather than bases, which contained magnesium or bases that were not made from magnesium. The formula of the compound identified was not always correct.

(ii) This equation proved challenging for many. Some answers incorrectly balanced equations which included all correct species. Use of an incorrect formula for magnesium nitrate was a common error, including Mg_3N_2.

(iii) Identification of the change in thermal stability down Group 2 nitrates was fairly well known.

5(a)(i)	$C_8H_{18} + 12½\ O_2 \rightarrow 8CO_2 + 9H_2O$ **M1** correct formula of M C_8H_{18} **M2** balanced equation showing complete combustion of M	**2**
5(b)(i)	**M1** ● high temperature ● high pressure ● N_2 Any two correct for **M1** **M2** $N_2 + O_2 \rightarrow 2NO$ **OR** $N_2 + 2O_2 \rightarrow 2NO_2$ **OR** $N_2 + xO_2 \rightarrow 2NO_x$	**2**
5(b)(ii)	**M1** NO_2 is the catalyst (for $SO_2 \rightarrow SO_3$) **OR** catalytic oxidation (of SO_2) by NO_2 **M2** $NO_2 + SO_2 \rightarrow NO + SO_3$ **AND** $2NO + O_2 \rightarrow 2NO_2$ **M3** SO_3 reacts with rain **OR** moisture/water (in atmosphere) to form sulfuric acid / H_2SO_4 **OR** $SO_3 + H_2O \rightarrow H_2SO_4$	**3**
5(b)(iii)	photochemical smog / PAN	**1**

Question 5

(a) (i) This was well answered by most candidates, although balancing the equation proved challenging for some.

(ii) Most candidates correctly mentioned the production of CO instead of CO_2. A very small number correctly referred to C and unburnt hydrocarbons.

(b) (i) A large proportion of candidates incorrectly interpreted this as a question about catalytic converters. Many answers used NO instead of N_2.

(ii) A small number of candidates correctly identified the catalytic role of NO_x and gave all the relevant equations. Many gave the equation for the formation of H_2SO_4, and a significant number gave equations and answers related to the formation of HNO_3 for acid rain.

(iii) This was well answered by most. Some candidates gave the identity of a gas instead of the type of air pollution.

(c) (i) Candidates often omitted dilute/(aq) with a named acid here.

(ii) A few candidates recognised that all three ester groups will undergo hydrolysis to form propanetriol. Many answers gave fragments where only one ester group had undergone hydrolysis.

(iii) This was well answered by most candidates. The majority of answers referred to esterification/substitution instead of condensation.

(iv) A small proportion of candidates answered this question correctly.

(v) A small proportion of candidates answered this question correctly.